

HNSS03 · Pharyngeal Arches — Development Timeline

Weeks post-conception, approximate · **EXAM** = flagged in class as testable · Origin key: **All layers** **Neural crest** **Paraxial / lateral plate mesoderm**

Endoderm (pouches) **Ectoderm (clefts)** **Arch 1** **Arch 2** **Arch 3** **Arches 4/6**

HoyaDocs

Updated Oct 1, 2026 · Rev 2

The Pharyngeal Arches				
WEEK	STAGE / STRUCTURE	WHAT HAPPENS	ORIGIN	DERIVATIVES / CLINICAL LINK
Wk 3–3.5	Arches 1 & 2 appear	Present: the three germ layers , pericardial bulge , cranial and caudal neuropores (still open) and arches 1 and 2 . Arches resemble gill formation, but no gill-like structures ever form	All layers	Step 1 simply says the apparatus forms in week 4 STEP 1
Wk 4	Arch 1 splits into two processes	Maxillary process (caudal to the frontal prominence, extends under the future nasal placode and eye): NC → premaxilla , maxilla , zygomatic , part of temporal by membrane ossification . Mandibular process (ventral, grows larger)	Arch 1 Neural crest	Arch 1 muscles (mastication, ant. digastric, mylohyoid, tensors) → CN V3
Wk 4–4.25	Arch 2 (hyoid arch)	Arises just after arch 1. Reichert's cartilage → stapes , styloid process, stylohyoid ligament, lesser horn + upper body of hyoid	Arch 2	Facial expression, stapedius, stylohyoid, post. digastric → CN VII
Wk 4.5	Meckel's cartilage	Meckel's cartilage populates the mandibular process. The mandible forms from the mesenchyme and NC around it — Meckel's is only a template	Arch 1 Neural crest	Trap: the mandible is not made from Meckel's cartilage
Wk 4.5–5	Neural crest migration	NC migrates from the hindbrain , midbrain and (less) forebrain; predetermined to form the facial skeleton . Rhombomeres: arch 1 = R1, R2 + midbrain · arch 2 = R4 · arch 3 = R6, R7 · arches 4/6 = R8 . NC carries the Hox code into each arch; SHH regulates Hox (SHH ^{-/-} mice are headless); OTX2 regulates arch 1	Neural crest	NC lacks SOD/catalase → alcohol (ROS) and retinoic acid (ROS + Hox) injure it; Tcof-1 → Treacher Collins
Wk 5	Arches 3 and 4/6 visible	Arch 3 → greater horn + lower body of hyoid. Arches 4/6 → laryngeal cartilages (thyroid, cricoid, arytenoid, corniculate, cuneiform) from lateral plate mesoderm , not NC EXAM . Arch 5 never forms	Arch 3 Arches 4/6	Arch 3: stylopharyngeus → IX · 4: superior laryngeal X · 6: recurrent laryngeal X
Wk 5–5.5	Arches grow	Arches grow significantly. Core mesenchyme is populated by paraxial + lateral plate mesoderm → muscles of face and neck; each arch has its own mesoderm → its own muscles	Mesoderm	"Crest builds the frame, mesoderm builds the motor"
Wk 5.5–6	Arches, pouches, clefts complete	All four cell types (mesenchyme, ectoderm, endoderm, NC) from three tissues form the arches (swellings), pouches (inside, endoderm) and clefts (outside, ectoderm). Each arch becomes separately innervated and gets its own artery	All layers	A muscle keeps its arch's nerve for life
Wk 8	Meckel's cartilage regresses	Meckel's disappears except two small dorsal portions → incus + malleus (endochondral ossification) EXAM	Arch 1	Malleus/incus = arch 1; stapes = arch 2

Sequence hook: "4 to arrive, 5 for the rest, 6 to get wired, 8 to shrink." Wk 4: arches 1–2 · wk 5: arches 3, 4/6 · wk 6: own nerve + artery · wk 8: Meckel's → malleus + incus.

★ **USMLE high-yield:** Nerve spine: **1 → V3** · **2 → VII** · **3 → IX** · **4 → X (SLN)** · **6 → X (RLN)**. Arch cartilage/bone = **neural crest** (except laryngeal cartilages = lateral plate mesoderm); arch muscle = **mesoderm**. All of this happens in the **embryonic period (wk 3–8)**, the window of greatest teratogen sensitivity.

The Pharyngeal Pouches (endoderm)

WEEK	STAGE / STRUCTURE	WHAT HAPPENS	ORIGIN	DERIVATIVES / CLINICAL LINK
Wk 5	Pouches 3 & 4 form wings	Endoderm at the distal ends forms dorsal and ventral wings . Pouch 3 dorsal → inferior parathyroid ; pouch 3 ventral → thymus ; pouch 4 dorsal → superior parathyroid	Endoderm	Pouches 3 + 4 fail → DiGeorge (22q11.2; Hoxa-3 knockout mimics it)
Wk 5–6	Pouch 1 → tubotympanic recess	Endoderm forms a diverticulum, the tubotympanic recess : distal → primitive tympanic cavity ; proximal → eustachian tube . The only pouch that contacts ectoderm (cleft 1)	Endoderm	Pouch 1 + cleft 1 → tympanic membrane
Wk 5–6	Pouch 2 buds	Epithelial endoderm proliferates into buds that penetrate the surrounding mesenchyme → palatine tonsils	Endoderm	—
Wk 5 → 10	Glands migrate	Pouch 3 glands lose their connection to the pharynx; the thymus migrates caudally and medially, pulling the inferior parathyroid with it, and fuses with its partner in the midline. The superior parathyroid attaches to the cranial thyroid and reaches its dorsal surface by wk 10	Endoderm	“3 takes the elevator down” → pouch 3 = inferior ; ectopic inferior parathyroid (thymus/mediastinum) STEP 1
With pouch 4	Pouch 5 → ultimobranchial body	Sometimes counted as part of pouch 4; forms the ultimobranchial body , which joins the thyroid → C cells → calcitonin	Endoderm	C cells → medullary thyroid carcinoma STEP 1
Month 4	Tonsillar crypt	The center of each pouch 2 bud undergoes apoptosis → hollow tonsillar crypt that fills with lymphoid tissue (definitive lymphoid tissue only in the last months of pregnancy)	Endoderm	—
→ Puberty	Thymus grows	Thymus keeps growing until puberty; in children it is large, behind the sternum, in front of the pericardium	Endoderm	Absent thymic shadow → DiGeorge

★ **USMLE high-yield:** “Ear, Tonsils, Bottom, Top”: **1** = middle ear + eustachian tube · **2** = palatine tonsil · **3** = inferior parathyroid + thymus · **4** = superior parathyroid · **5** = C cells. Glands come from **pouches**, never arches.

The Pharyngeal Clefts (ectoderm)

WEEK	STAGE / STRUCTURE	WHAT HAPPENS	ORIGIN	DERIVATIVES / CLINICAL LINK
Wk 5–6	Cleft 1 → ear canal	The dorsal part of cleft 1 penetrates the mesoderm → external auditory meatus — the only definitive cleft structure	Ectoderm	“1 keeps its ear; the rest disappear”
Wk 5–7	Cleft 2 overgrowth → cervical sinus	Cleft 2 grows caudally over clefts 3, 4 and 5 (textbooks: “arch 2 overgrows arches 3–4”) and fuses with the epicardial ridge , enclosing the cervical sinus , which is normally obliterated	Ectoderm Arch 2	Sinus persists → branchial cleft cyst (lateral, anterior to SCM) ± fistula

★ **USMLE high-yield:** **Lateral** neck cyst anterior to the SCM = branchial cleft cyst (cleft 2). **Midline** cyst that moves with swallowing = thyroglossal duct cyst (HNSS04). Cleft and pouch timing beyond the lecture's stated weeks are approximate.